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Low-Complexity Geometry-Aware Hybrid Beamforming for Near-Field STAR-RIS Networks

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Presented By: name of presenter

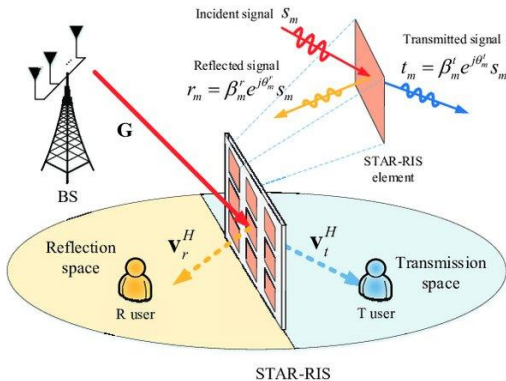
1. Context & Motivation

STAR-RIS: A simultaneously transmitting and reflecting surface that covers both sides of space. Part of the incident energy is reflected, part is transmitted. Users only receive; T-side or R-side is decided by location.

Hybrid beamforming: The base station uses a limited number of RF chains to serve several users at once. Analog plus digital precoding keeps hardware cost manageable, so only a few streams can be scheduled in each slot.

Constraint: Large STAR-RIS, near-field spherical-wave channels, and one common transmit/reflect profile per side couple user selection, surface configuration, and hybrid beams.

2. Overall Architecture



STAR-RIS architecture

- **Limited hybrid stream budget**
Few RF chains, so only a small set of users can be served in one slot, with a cap on each STAR-RIS side.
- **PF-correlation user selection**
Rank users by a long-term fairness score, then drop highly correlated users so the scheduled set stays spatially compatible.
- **Shared STAR-RIS profile per side**
Same-side users share one transmit or reflect profile and one energy split; the surface cannot be tuned per user.

3. System Model

Overview. A limited stream budget requires a small S ; PF-correlation scoring forms that set without locking onto the strongest users. A shared STAR-RIS profile then forces one (β, ν) per side before hybrid precoding.

1. G — near-field map

$[G]_{n,m}$ from spherical distances.

Builds the geometry later stages use.

2. PF — fairness rank

$$\psi_k = \widehat{G}_k / (T_k + \varepsilon).$$

Ranks users by long-term fairness, not only instantaneous gain.

3. S — scheduled set

$|S| \leq N_s$ and side caps.
A set the hybrid BS can serve.

6. F — hybrid precoder

$$F = F_{\text{RF}} F_{\text{BB}}, \|F\|_F^2 \leq P.$$

Separates the scheduled users on the already-shaped channels.

5. ν — common profiles

$\nu_k = \nu_T$ or ν_R , then form h_k .

One surface setting per side; F does the per-user split.

4. β — side energy split

$\beta_T + \beta_R = 1$ from T/R demand of S .

Shares the surface energy between the two sides of the chosen set.

4. Proposed GA-HBF-PFBDP Scheme

Approach. Rather than alternating over all RIS coefficients and beams, the six blocks run in one pass: PF-BDP picks a feasible S , then a common (β, v_T, v_R) is built from geometry, and F is formed on the resulting channels. Input: $G, \{r_k\}, \{T_k\}, N_s, N_T^{\max}, N_R^{\max}, P$. Output: S, v_T, v_R, F .

1. Scores

Compute $\hat{G}_k, \psi_k$, and $\rho_{i,j}$.
Turns geometry into a PF rank and a correlation map.

2. Select S

Maximize PF minus correlation.
Under $|S| \leq N_s$ and side caps.

3. Energy split

Set (β_T, β_R) from T/R demand of S .
Puts more energy on the side that S actually uses.

6. Hybrid F

Phase F_{RF} , RZF F_{BB} , then normalize.
Analog and baseband beams for the chosen users.

5. Effective h_k

$h_k^H = r_k^T \text{diag}(v_k) G, k \in S$.
Cascaded channels after the common profiles are applied.

4. Profiles

Build common v_T, v_R from geometry.
One TX and one RX setting for the whole surface.

5. Simulation Results (Rate)

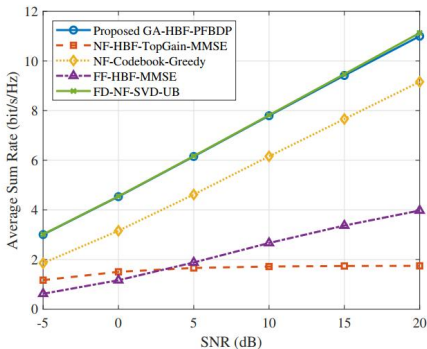


Figure 1. Average sum-rate versus SNR for different beamforming schemes in a large STAR-RIS near-field system.

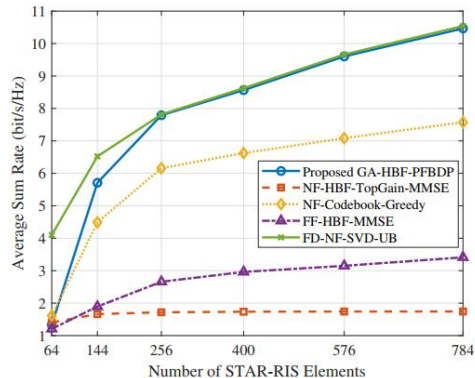


Figure 2. Average sum-rate versus the number of STAR-RIS elements for different beamforming schemes.

6. Simulation Results (Runtime and Users)

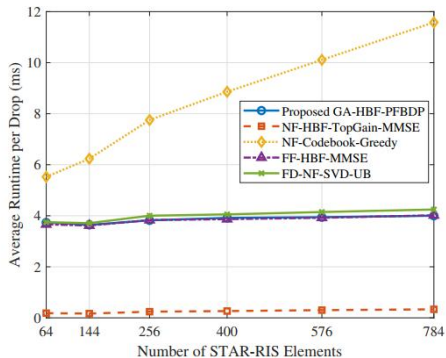


Figure 3. Average runtime per drop versus the number of STAR-RIS elements for different beamforming schemes.

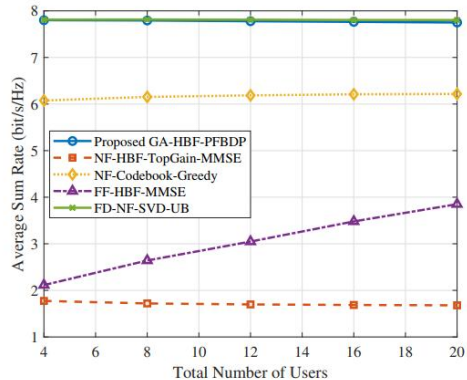


Figure 4. Average sum-rate versus the total number of users for different beamforming schemes.

7. References

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Thank You

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